

Electronics Device & Circuits

EE 223

Reference material for Analog Electronics

by

Dr. T. S. Bheemraj

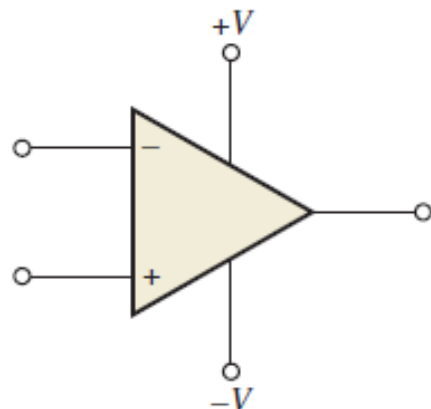
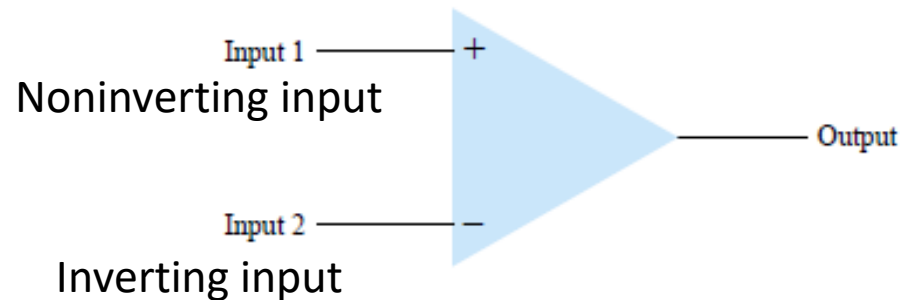
Assistant Professor, EED, MANIT Bhopal

Outcome 1: To understand basics of OP AMPs

- What is an op-amp or operational amplifier?
- A very **high gain differential amplifier** with **high input impedance** (typically a few megohms) and **low output impedance** (less than 100).
- Where are op-amps used?
- To provide voltage amplitude changes (subtraction, addition, differentiator, integrator), oscillators(Wien Bridge Oscillator, RC Phase Shift Oscillator), active filters (low pass, high pass, band pass) and instrumentation circuits such as dc or ac voltmeters.

Outcome 1: To understand basics of OP AMPs

- What is the symbol of an op-amp? Or How does an op-amp look like?



Symbol with dc supply connections

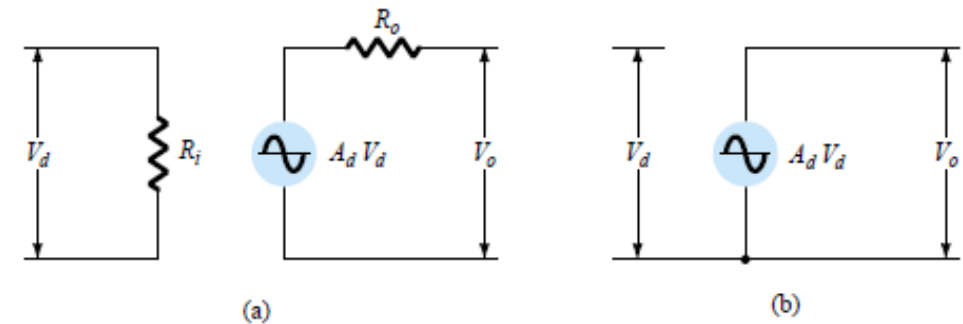


Figure 14.11 Ac equivalent of op-amp circuit: (a) practical; (b) ideal.

Voltage controlled voltage source studied in circuits and network

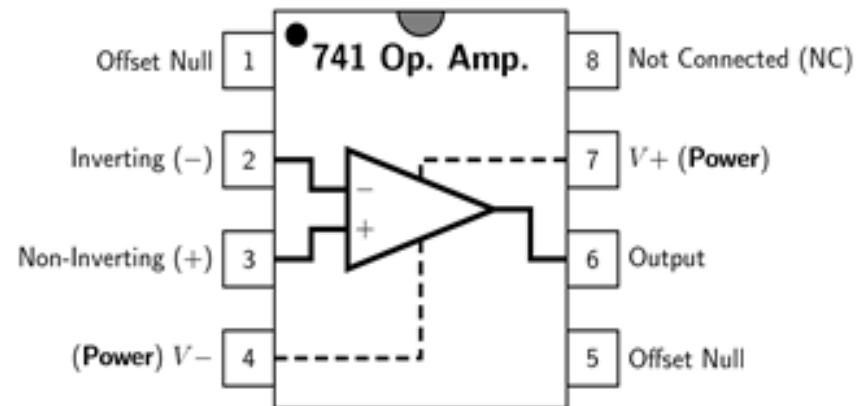
Advantage of high input impedance: entire input voltage is amplified

Advantage of low input impedance: no voltage is dropped

Ideal op-amp has *infinite voltage gain*, *infinite bandwidth*, *infinite input impedance* and *zero output impedance*.

Outcome 1: To understand basics of OP AMPs

- How does a practical op-amp look like?



Ceramic is used in high-reliability applications, while plastic is used in commercial electronics.

Temperature: Ceramic can withstand high temperatures, while plastic may melt or deform.

Durability: Plastic is more durable than ceramic.

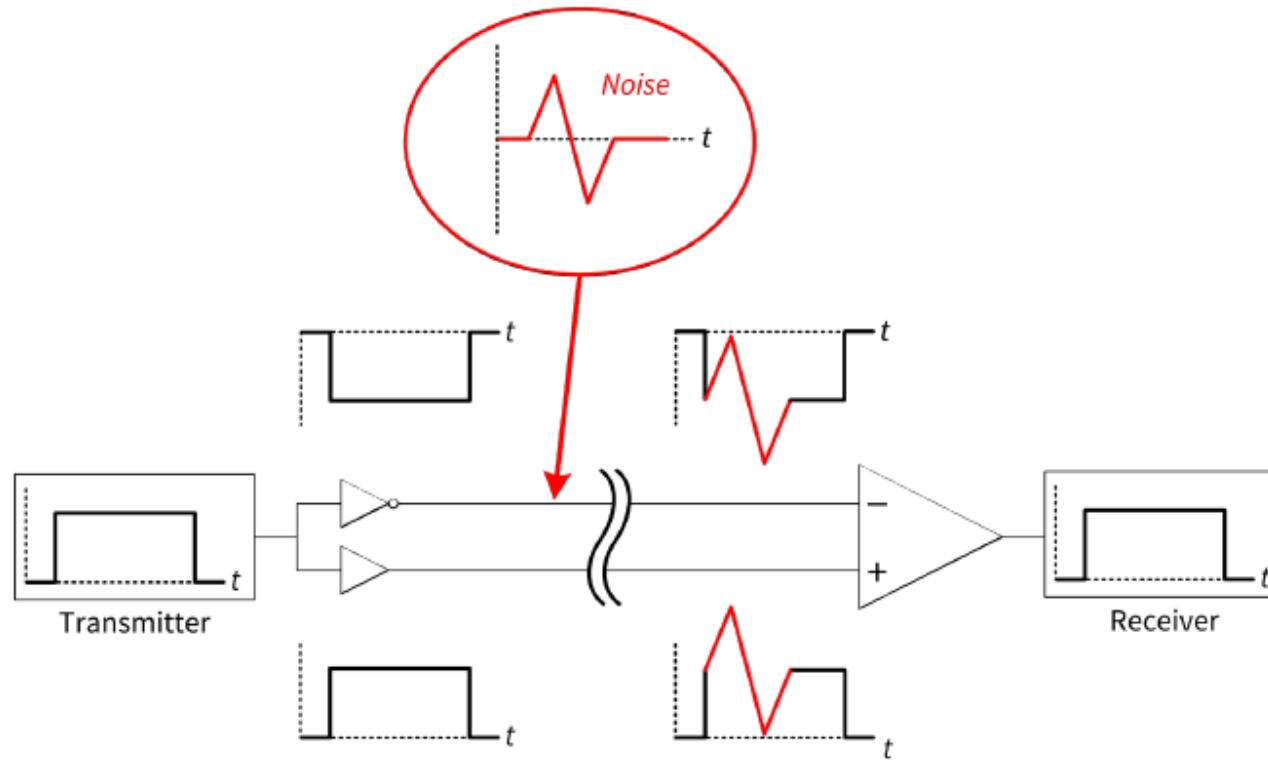
Practically an op-amp is fabricated in an IC. For example LM741 is an 8 PIN IC with package 8-Pin ceramic dual in-line package (CDIP) or plastic dual in-line package (PDIP).

This IC will be used in your lab session

Outcome 1: To understand basics of OP AMPs

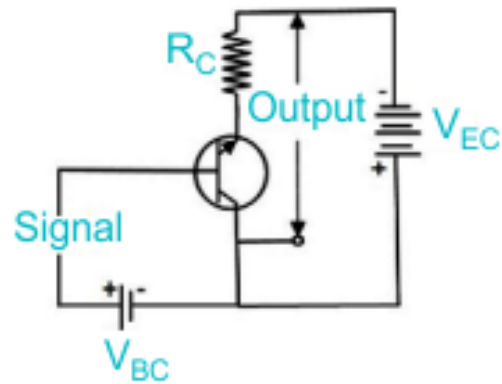
- What is a differential amplifier?
- A differential (or difference) amplifier is a two-input circuit that amplifies only the difference between its two inputs. An operational amplifier or op-amp is an example of a difference amplifier.
- **Why differential amplifiers are used ?**
- Differential amplifiers are useful in electrically noisy environments where a low amplitude electrical signal can be easily corrupted by the effect of unwanted external noise. In this scenario, a single-ended amplifier would be unsuitable since it would also amplify the unwanted noise signal as well as the desired input signal. A differential amplifier works on the principle that unwanted electrical noise couples equally onto both input terminals of the amplifier and will therefore be rejected allowing only the wanted signal to be amplified.

Outcome 2



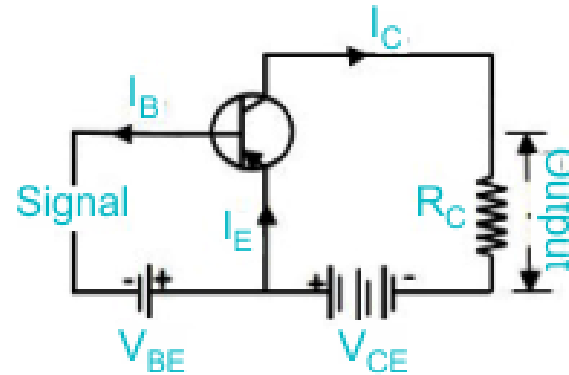
Gate driver or send signals to longer distance

Outcome 1: To understand basics of OP AMPs



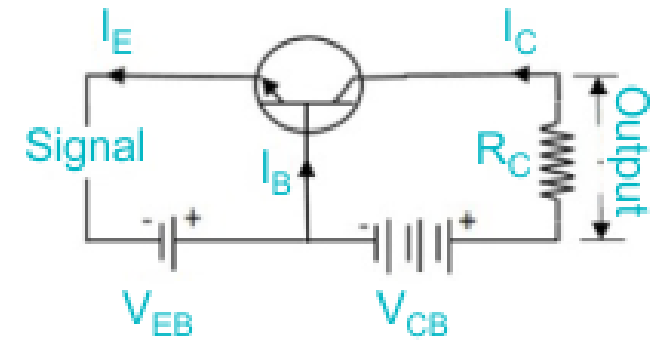
Using NPN Transistor

Common Collector Configuration



Using NPN transistor

Common Emitter Configuration



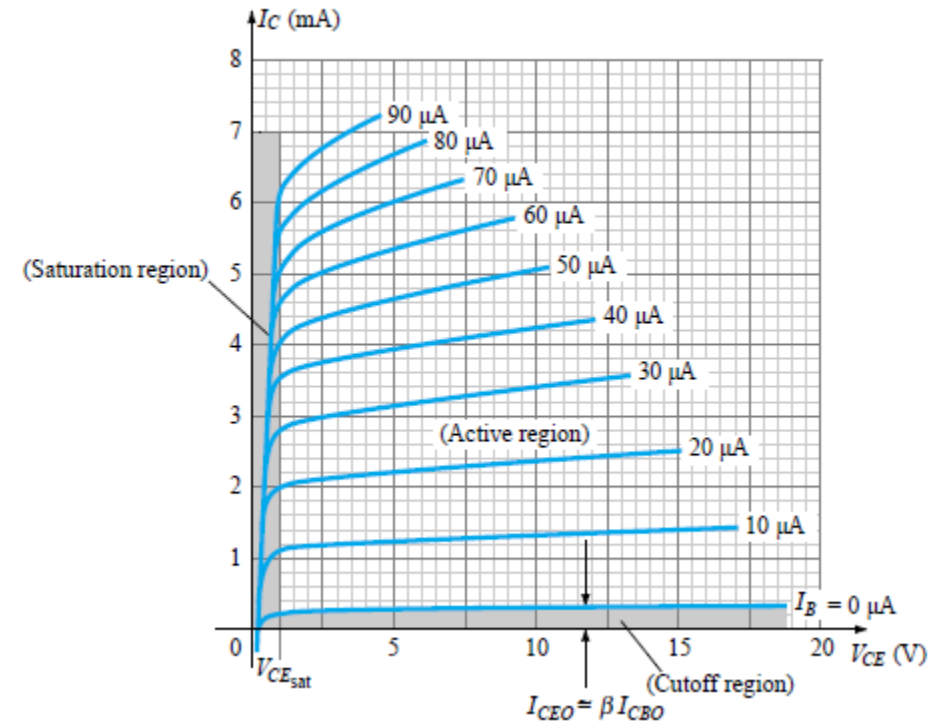
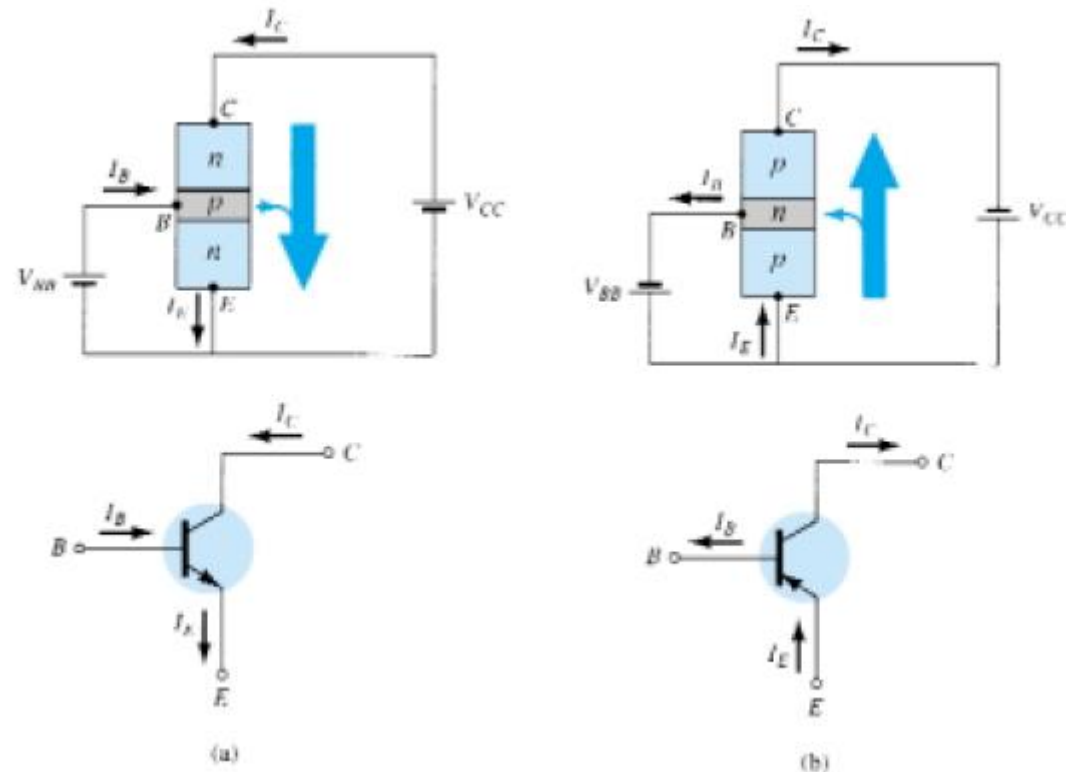
Using NPN Transistor

Common Base Configuration

NPN transistors are more popular than PNP transistors because electrons, the majority charge carriers in NPN transistors, have significantly higher mobility than holes (the majority carriers in PNP transistors), resulting in faster switching speeds, better conductivity, and overall superior performance in most electronic circuits.

Outcome 1: To understand basics of OP AMPs

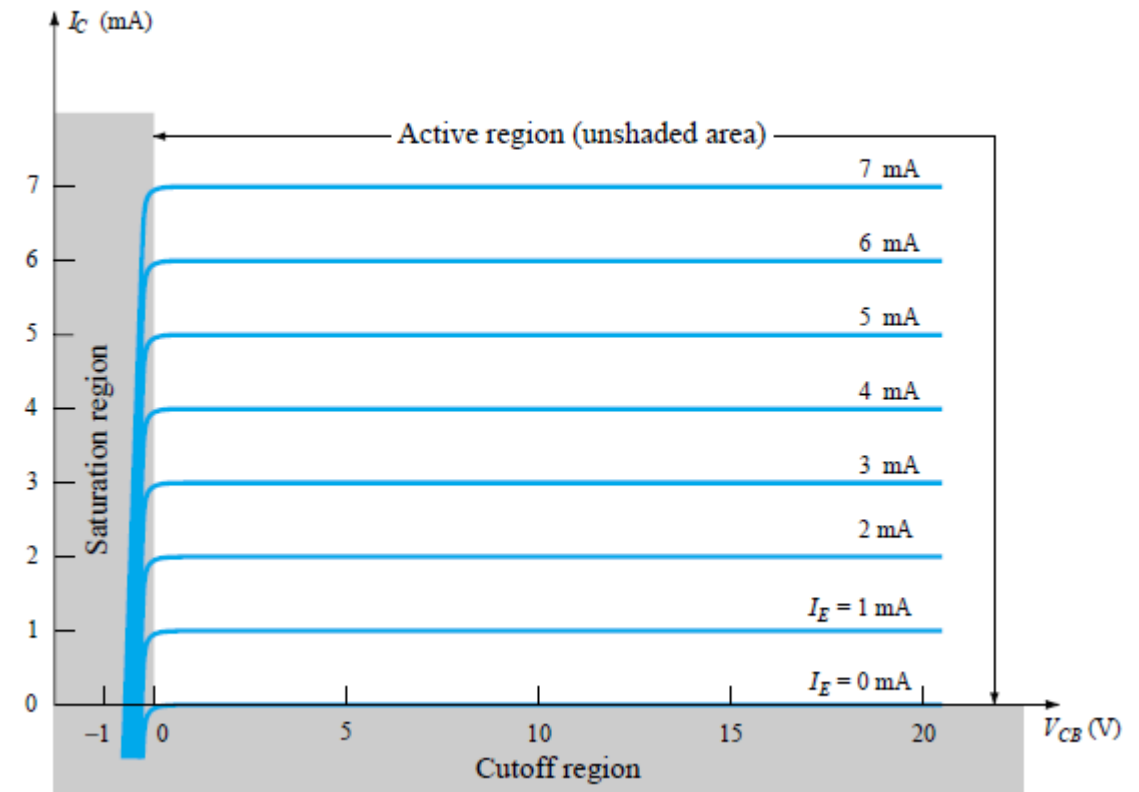
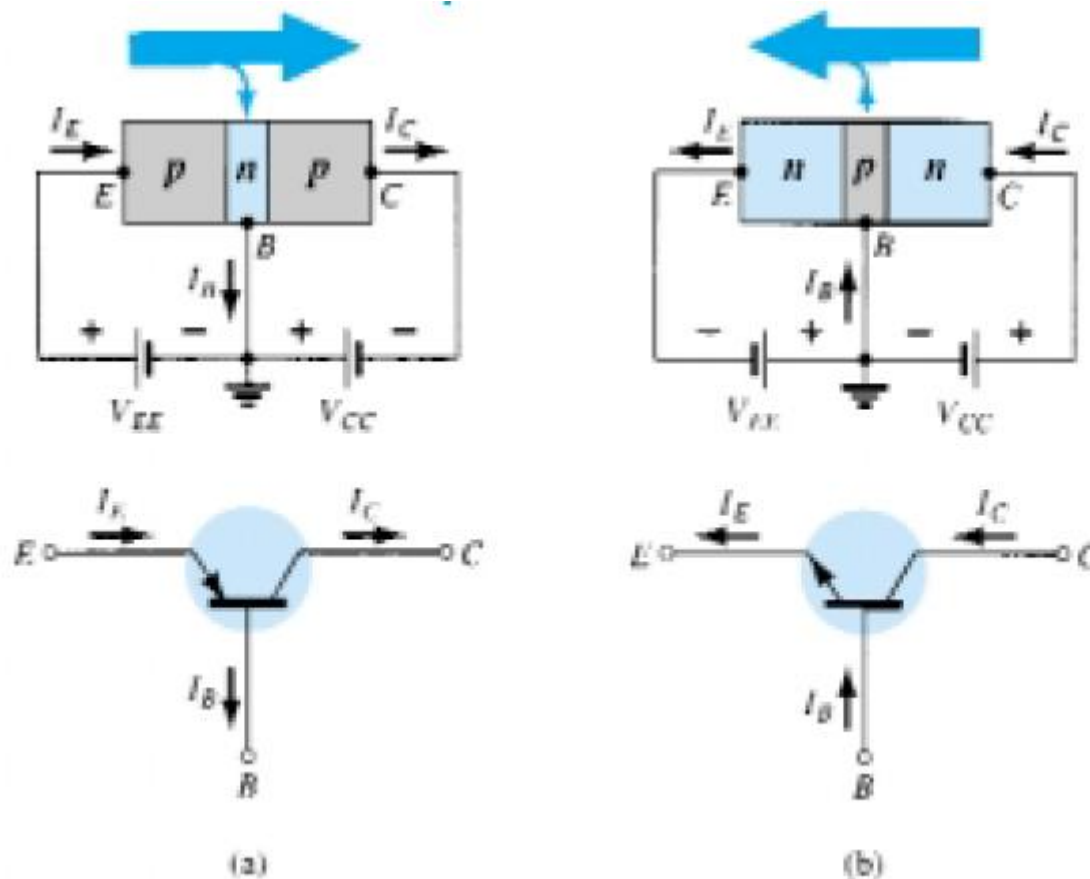
- Common emitter configuration



In the active region of a common-emitter amplifier the collector-base junction is reverse-biased, while the base-emitter junction is forward-biased.

Outcome 1: To understand basics of OP AMPs

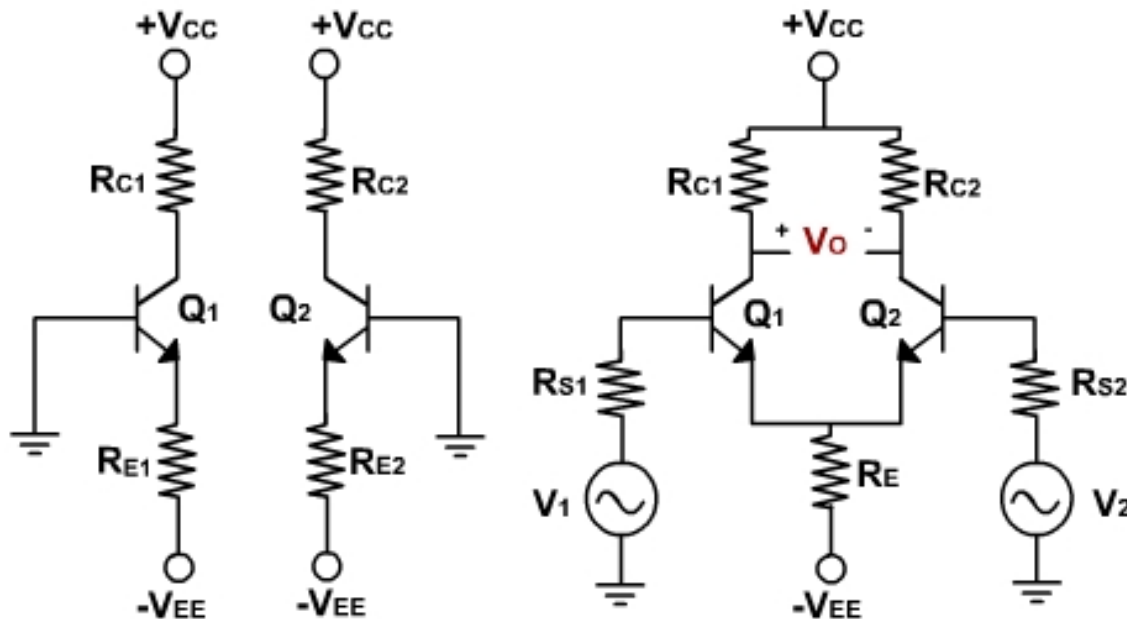
- Common Base configuration



In the active region the collector-base junction is reverse-biased, while the base-emitter junction is forward-biased.

Outcome 1: To understand basics of OP AMPs

- How the differential amplifier is developed?



In common emitter configuration

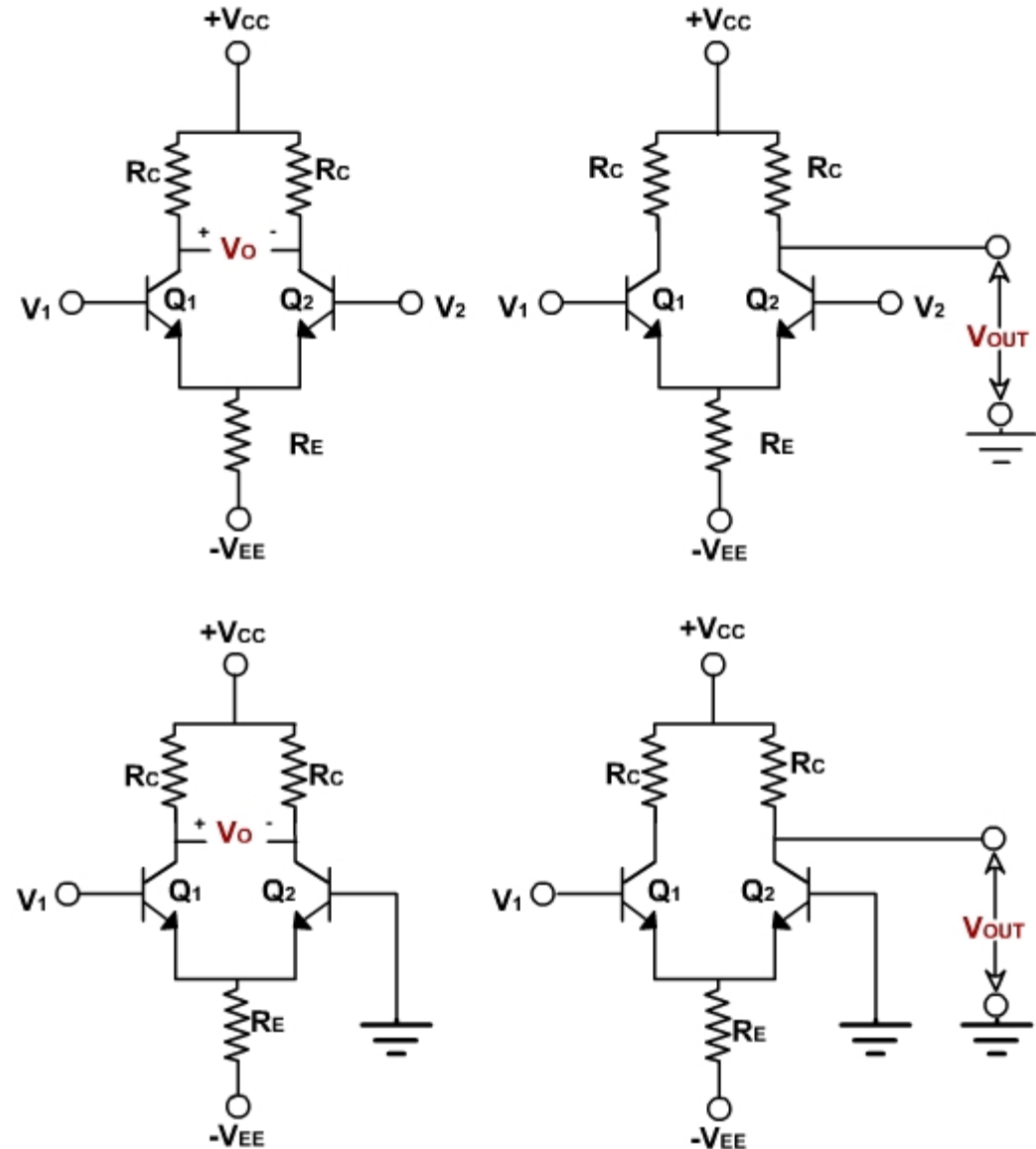
$$i_c = \beta \times i_B$$

$$I_e = I_B + I_C$$

Ideally, the output voltage is zero when the two inputs are equal. When v_1 is greater than v_2 the output voltage with the polarity shown appears. When v_1 is less than v_2 , the output voltage has the opposite polarity.

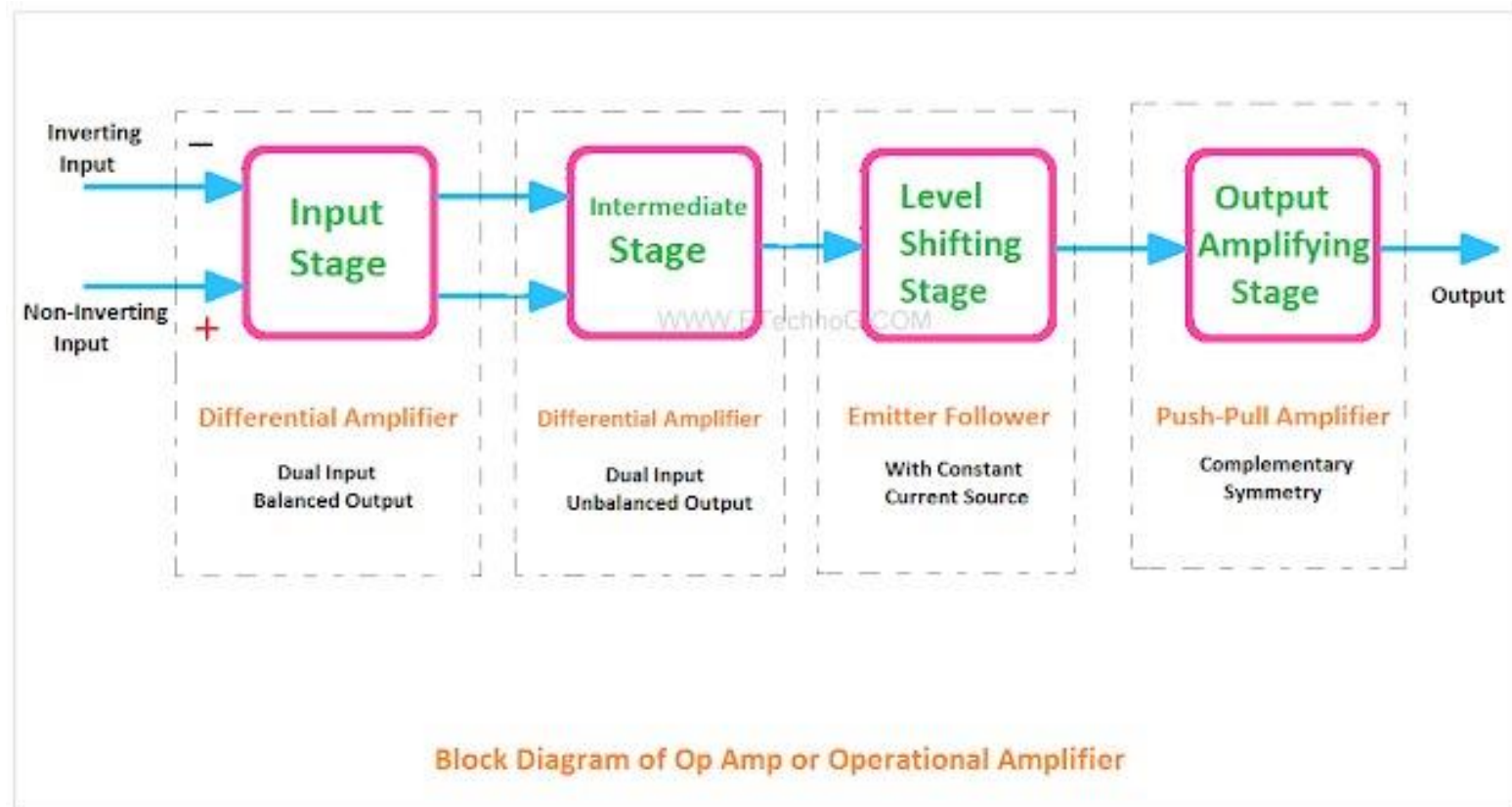
Outcome 1: To understand basics of OP AMPs

- Types of differential amplifier
- Dual input, balanced output differential amplifier.
- Dual input, unbalanced output differential amplifier.
- Single input balanced output differential amplifier.
- Single input unbalanced output differential amplifier.



Outcome 1: To understand basics of OP AMPs

- Op-amp block diagram



Outcome 1: To understand basics of OP AMPs

- Types of differential amplifier
- Dual input, balanced output differential amplifier or Differential input differential output
- Dual input, unbalanced output differential amplifier or Differential input single ended output
- Single input balanced output differential amplifier or single ended input differential output
- Single input unbalanced output differential amplifier or single ended input single ended output

Outcome 1: To understand basics of OP AMPs

- $V_{out} = V_{o1} - V_{o2}$, Whereas $V_{o1} = AV_1$ and $V_{o2} = AV_2$
- Thus $V_{out} = A(V_1 - V_2)$
- Input V_1 is called non-inverting input as V_{out} is in phase with V_1
- Input V_2 is called inverting input as V_{out} is out of phase with V_2
- Input bias current is average of DC base current
- Input offset current is difference of DC base current
- Output offset voltage is voltage observed at output when no input
- Input offset voltage is voltage to be provided at input to reduce output offset voltage to zero

Outcome 1: To understand basics of OP AMPs

- Common Mode Rejection Ratio
- When two signal voltages of the same phase, frequency, and amplitude are applied to the two inputs of op-amp, they tend to cancel, result in a zero output voltage. This action is called *common-mode rejection*.
- Desired signals can appear on only one input or with opposite polarities on both input lines. Whereas Unwanted signals (noise) appears with the same polarity on both input lines.
- The measure of an amplifier's ability to reject Unwanted signals (common-mode) signals is a parameter called the **CMRR (common-mode rejection ratio)** represented in dB as shown below.

$$\text{CMRR} = 20 \log \left(\frac{A_{ol}}{A_{cm}} \right)$$

Open Loop Gain of IC 741 is 2×10^5 V/V

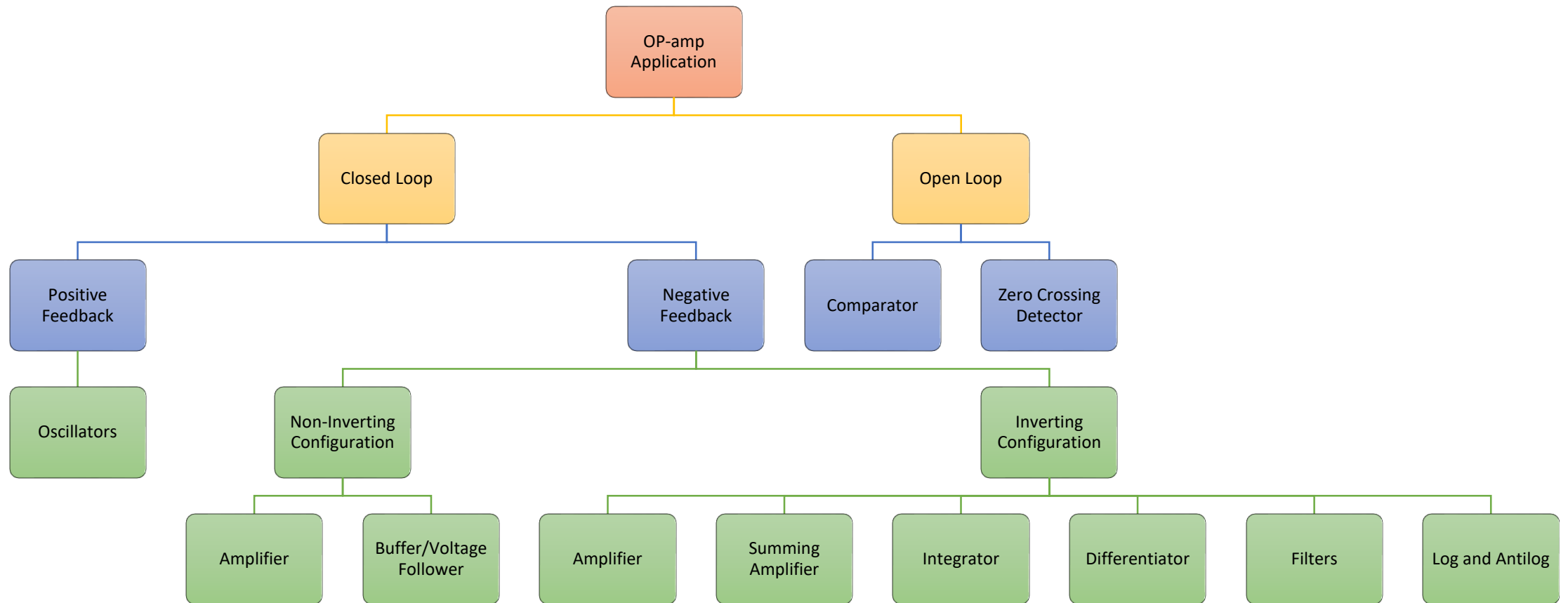
Outcome 1: To understand basics of OP AMPs

- Slew Rate
- The maximum rate of change of the output voltage in response to a step input voltage is the slew rate of an op-amp. The unit of slew rate is volts per microsecond (V/μs).

$$\text{Slew rate} = \frac{\Delta V_{out}}{\Delta t}$$

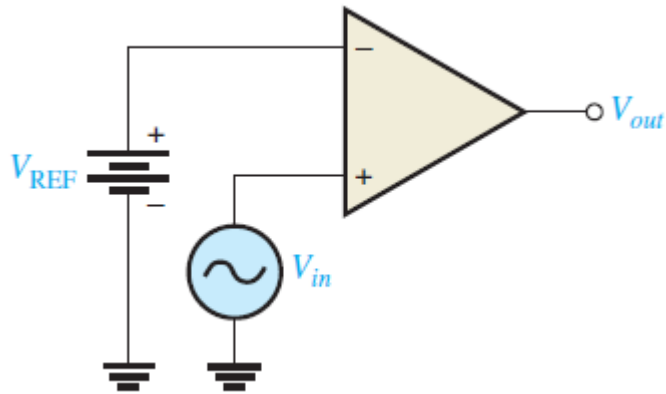
- If the rate of voltage change is greater than the slew rate, the output would not be able to change fast enough and would not vary over the full range expected, resulting in signal clipping or distortion.

Outcome 1: To understand basics of OP AMPs

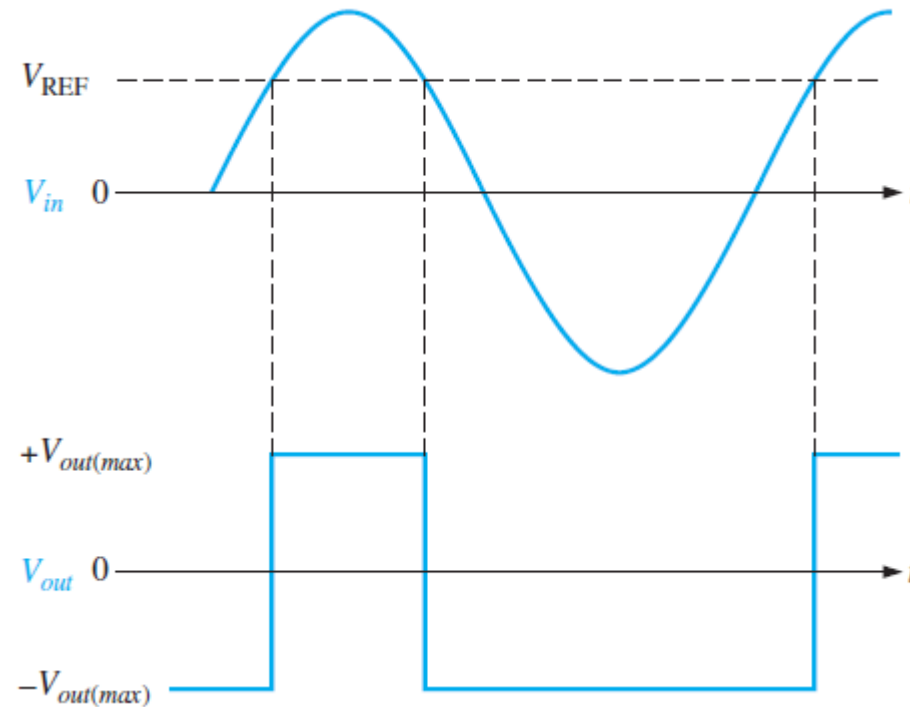


Outcome 1: To understand basics of OP AMPs

Comparator

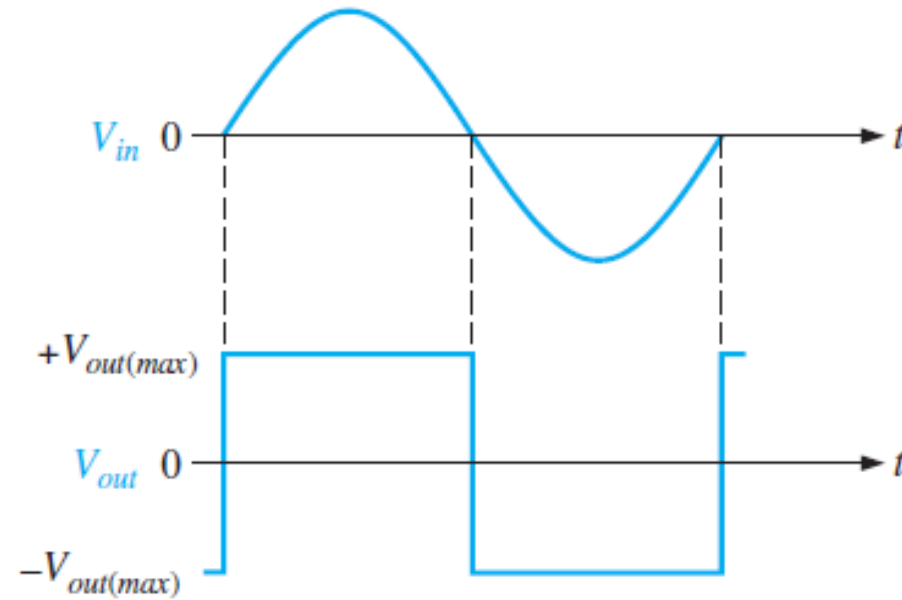
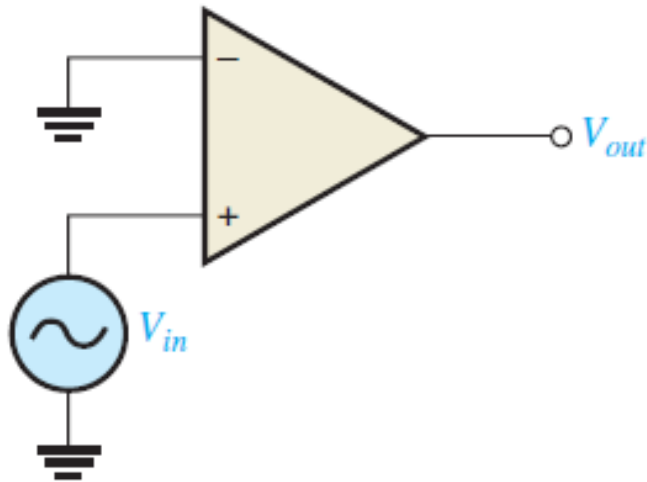


If V_{ref} is greater than V_{in} , V_{out} is $+V_{cc}$
If V_{ref} is less than V_{in} , V_{out} is $-V_{cc}$



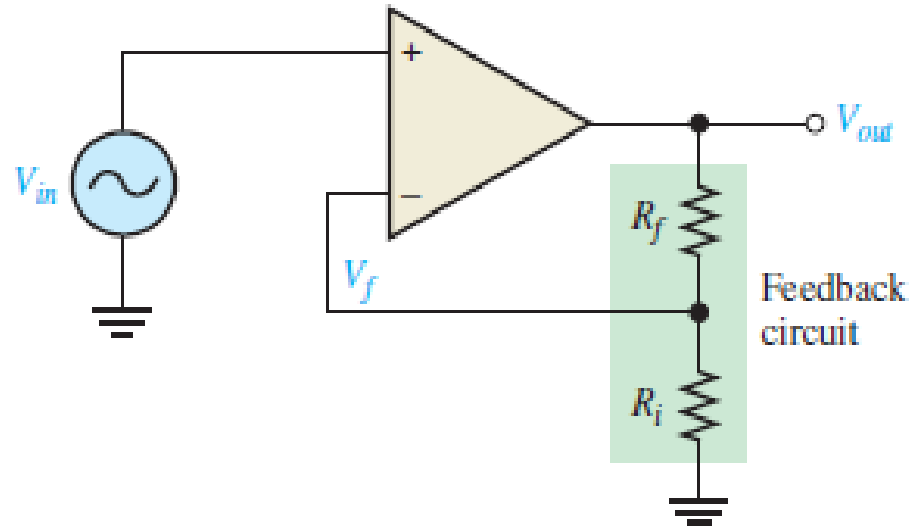
Outcome 1: To understand basics of OP AMPs

Zero Crossing Detector



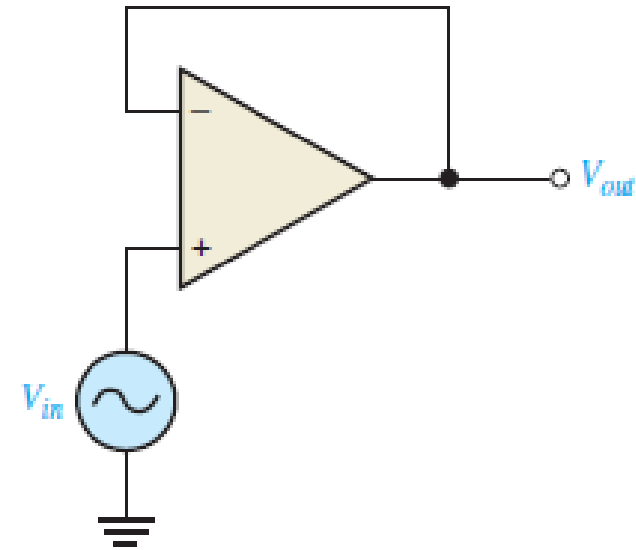
Outcome 1: To understand basics of OP AMPs

Non-Inverting Amplifier



$$V_{out} = \left(1 + \frac{R_f}{R_i}\right) V_{in}$$

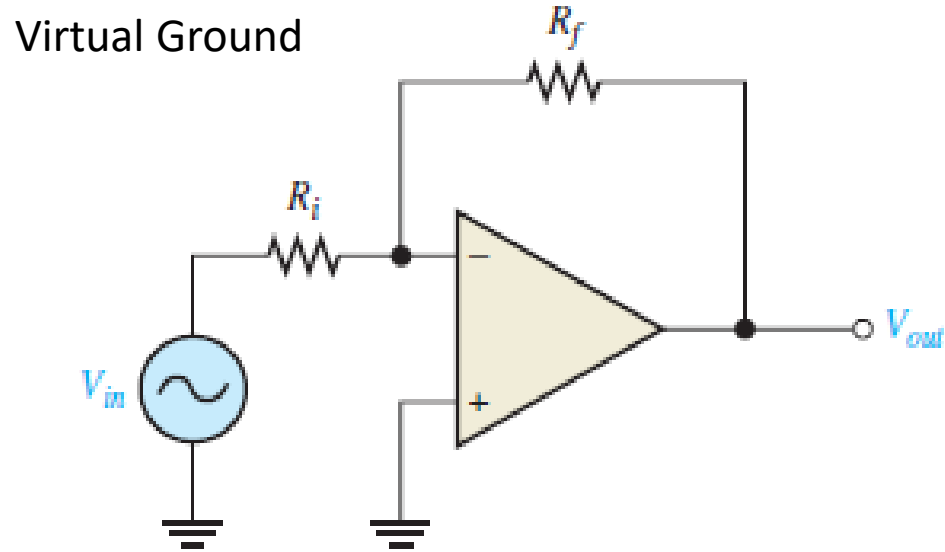
Voltage Follower/ Buffer



$$V_{out} = V_{in}$$

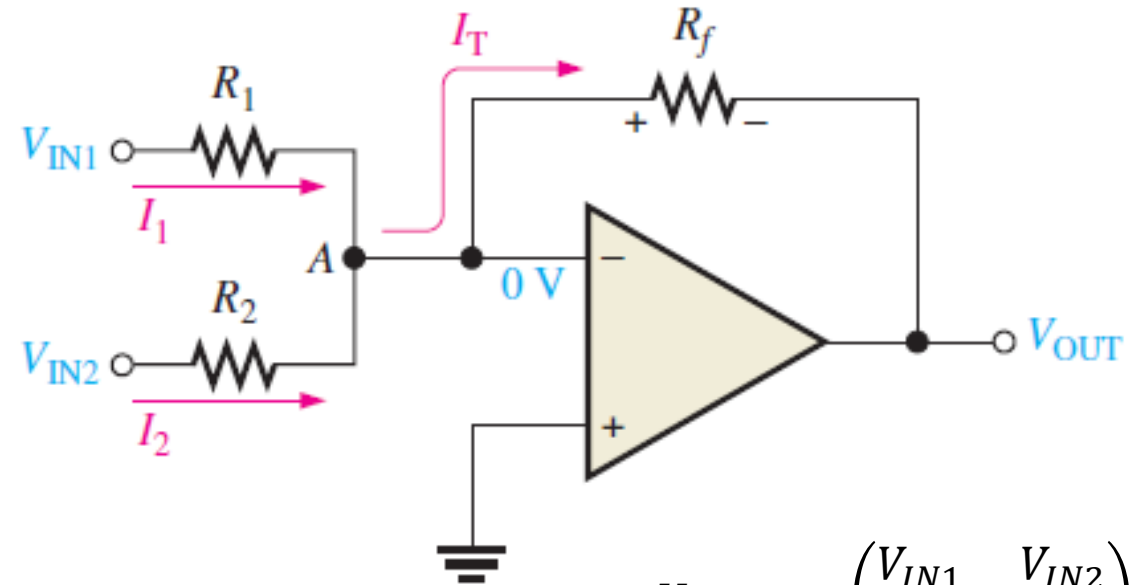
Outcome 1: To understand basics of OP AMPs

Inverting Amplifier



$$V_{out} = -\left(\frac{R_f}{R_i}\right)V_{in}$$

Summing Amplifier



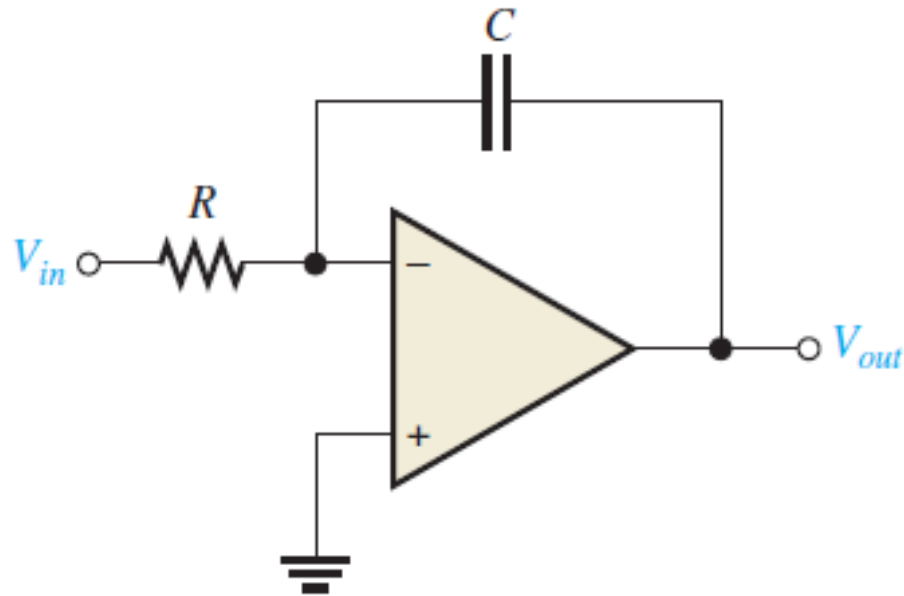
$$V_{out} = -\left(\frac{V_{IN1}}{R_1} + \frac{V_{IN2}}{R_2}\right)R_f$$

If all three of the resistors are equal ($R_1 = R_2 = R_f = R$), then

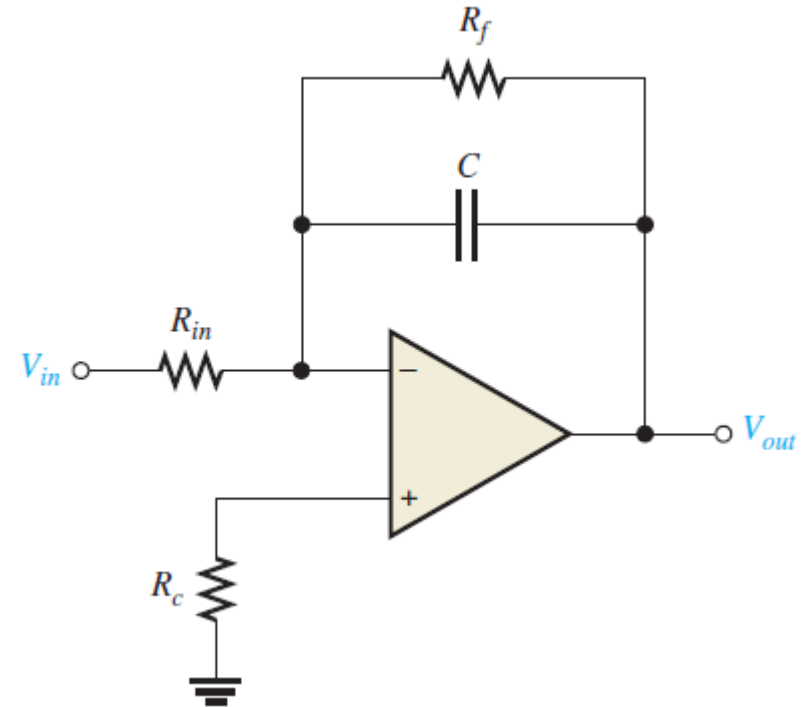
$$V_{out} = -(V_{in1} + V_{in2})$$

Outcome 1: To understand basics of OP AMPs

Integrator



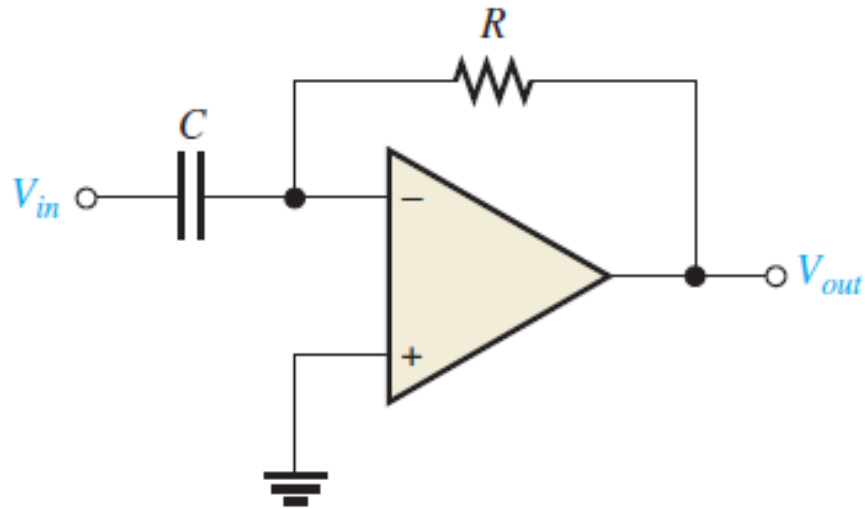
$$V_{out}(t) = -\frac{1}{RC} \int V_{in}(t) dt$$



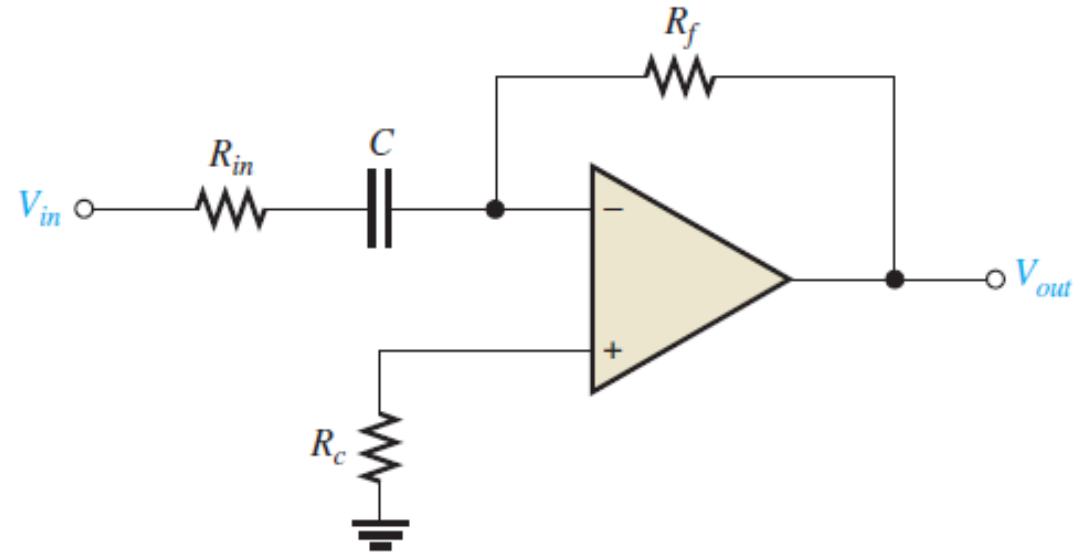
Practical Integrator

Outcome 1: To understand basics of OP AMPs

Differentiator



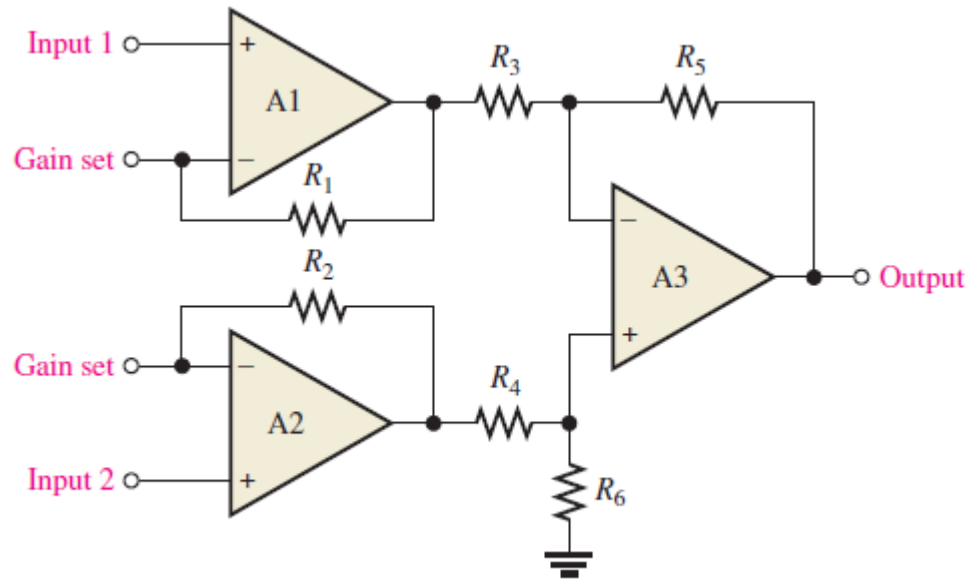
$$V_{out}(t) = -RC \frac{dV_{in}(t)}{dt}$$



Practical Differentiator

Outcome 1: To understand basics of OP AMPs

instrumentation amplifier



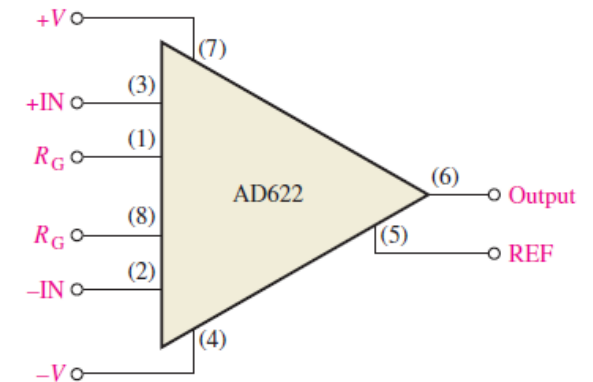
$$R3 = R5 = R4 = R6$$

$$R1 = R2 = R$$

The main purpose of an instrumentation amplifier is to amplify small signals that may be riding on large common mode voltages.

► FIGURE 14-4

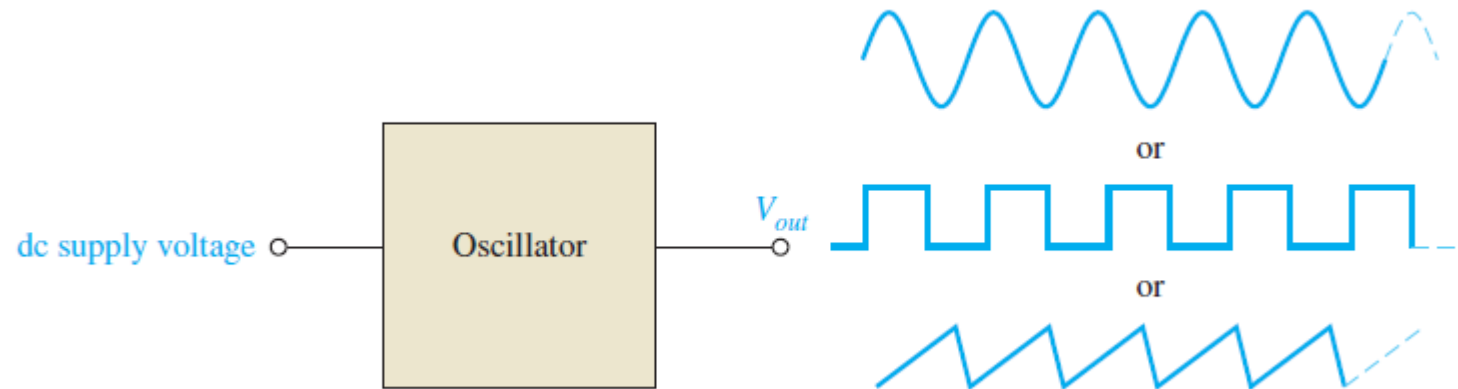
The AD622 instrumentation amplifier.



Outcome 1: To understand basics of OP AMPs

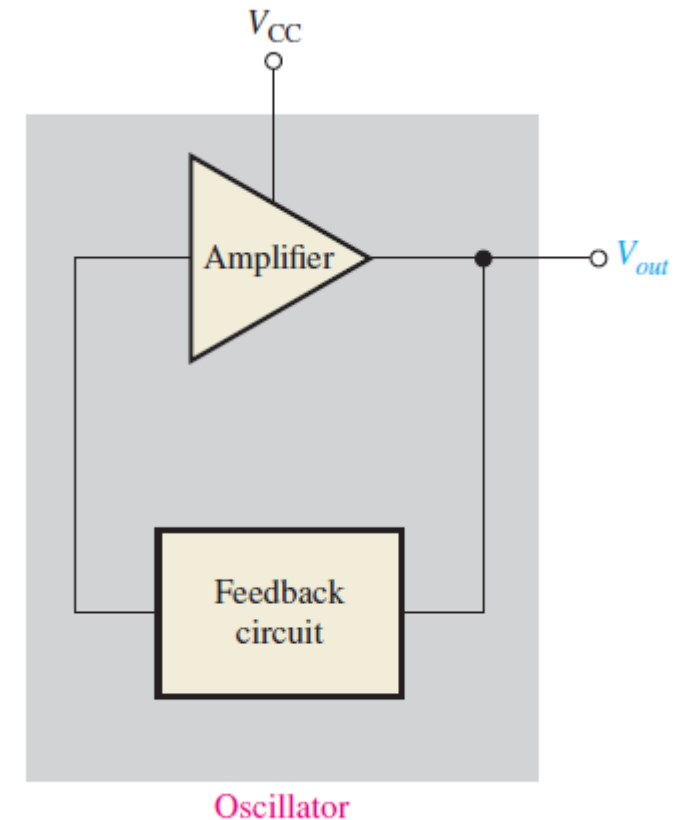
- Oscillators are electronic circuits that generate an output signal without the necessity of an input signal.
- Different types of oscillators produce various types of outputs including sine waves, square waves, triangular waves, and sawtooth waves.
- Sinusoidal oscillator operation is based on the principle of positive feedback, where a portion of the output signal is fed back to the input in a way that causes it to reinforce itself and thus sustain a continuous output signal.

Outcome 1: To understand basics of OP AMPs

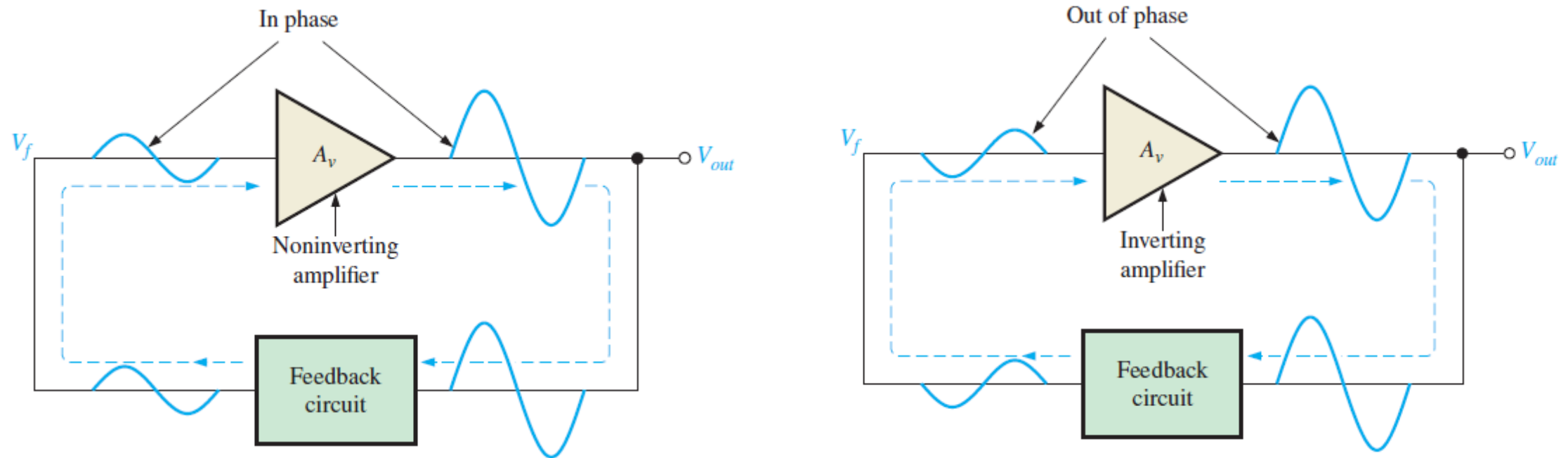


Feedback Oscillators One type of oscillator is the **feedback oscillator**, which returns a fraction of the output signal to the input with no net phase shift, resulting in a reinforcement of the output signal.

Relaxation Oscillators A second type of oscillator is the **relaxation oscillator**. Instead of feedback, a relaxation oscillator uses an RC timing circuit to generate a waveform that is generally a square wave or other nonsinusoidal waveform. Typically, a relaxation oscillator uses a Schmitt trigger or other device that changes states to alternately charge and discharge a capacitor through a resistor. **555 timer**



Outcome 1: To understand basics of OP AMPs



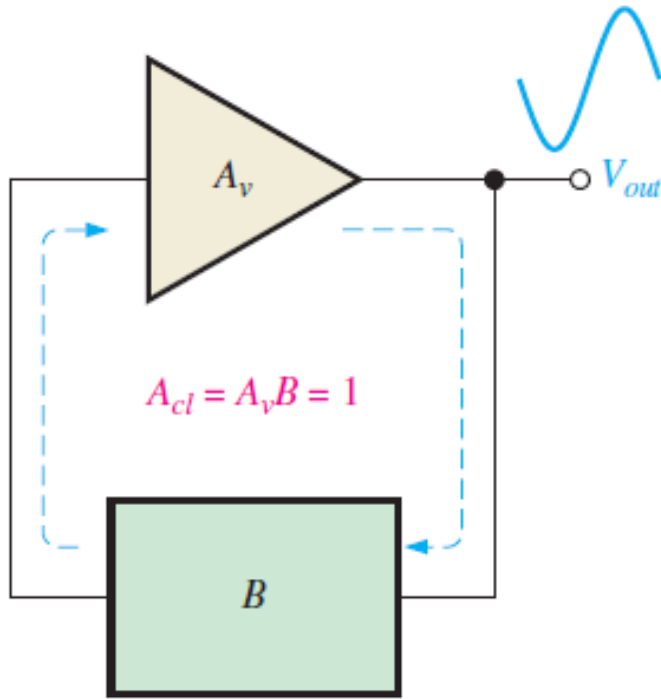
Positive feedback is characterized by the condition wherein a portion of the output voltage of an amplifier is fed back to the input with no net phase shift around the loop, resulting in a reinforcement of the output signal.

Outcome 1: To understand basics of OP AMPs

- **Conditions for Oscillation**

- The phase shift around the feedback loop must be effectively 0°. The feedback circuit accomplishes the necessary phase shift. For a noninverting amplifier the input to the amplifier is returned in phase with the output. For an inverting amplifier, the input is returned 180° out of phase with the output.
- The voltage gain, A_{cl} , around the closed feedback loop (loop gain) must equal or greater than 1 (unity). For a sine wave oscillator, the loop gain must be exactly 1; otherwise the output is distorted with clipping.

Outcome 1: To understand basics of OP AMPs

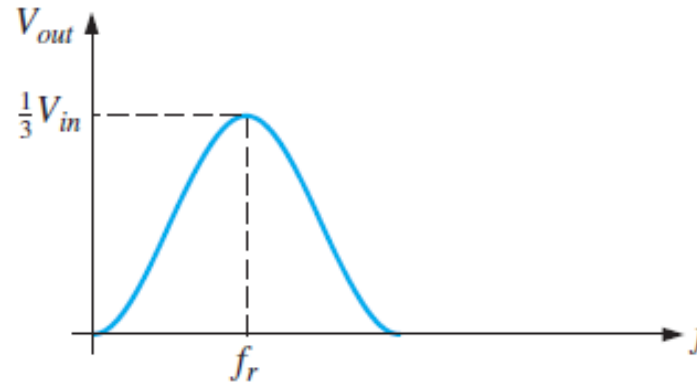
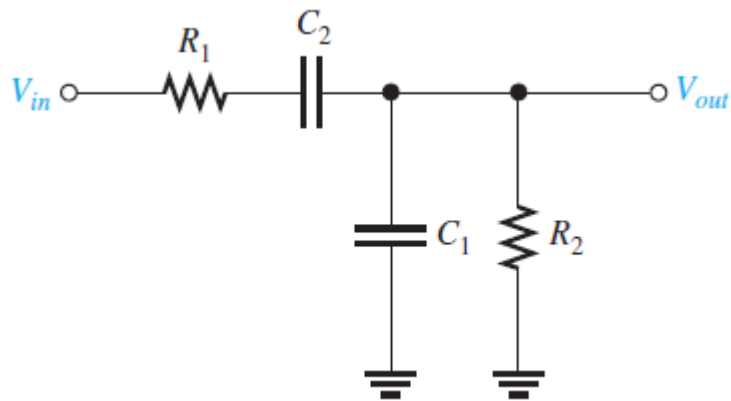


The voltage gain around the closed feedback loop, A_{cl} , is the product of the amplifier gain, A_v , and the attenuation, B , of the feedback circuit.

$$A_{cl} = A_v B$$

Outcome 1: To understand basics of OP AMPs

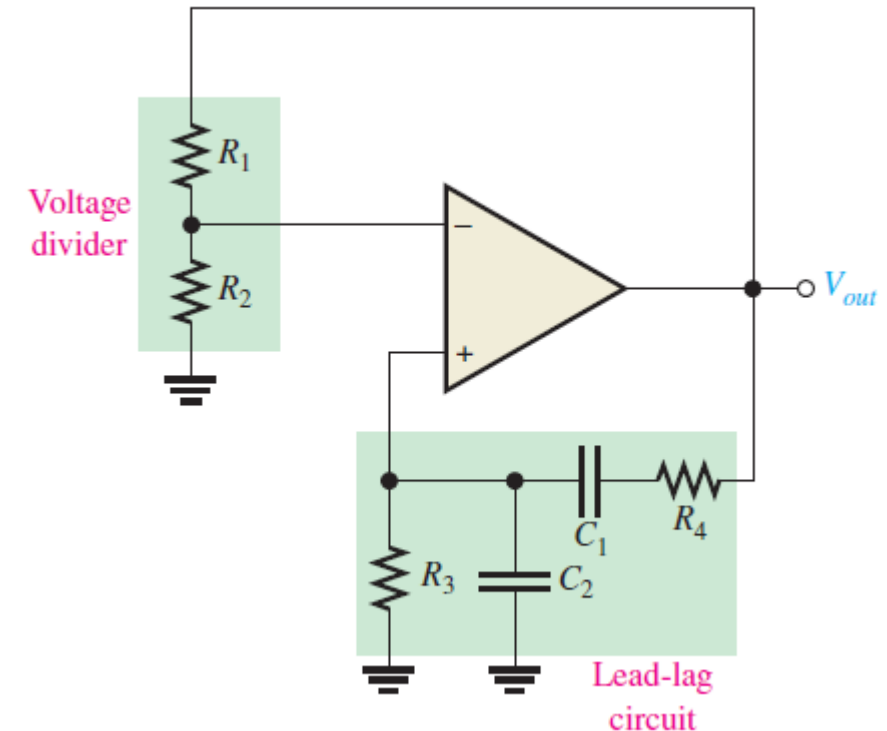
The Wien-Bridge Oscillator



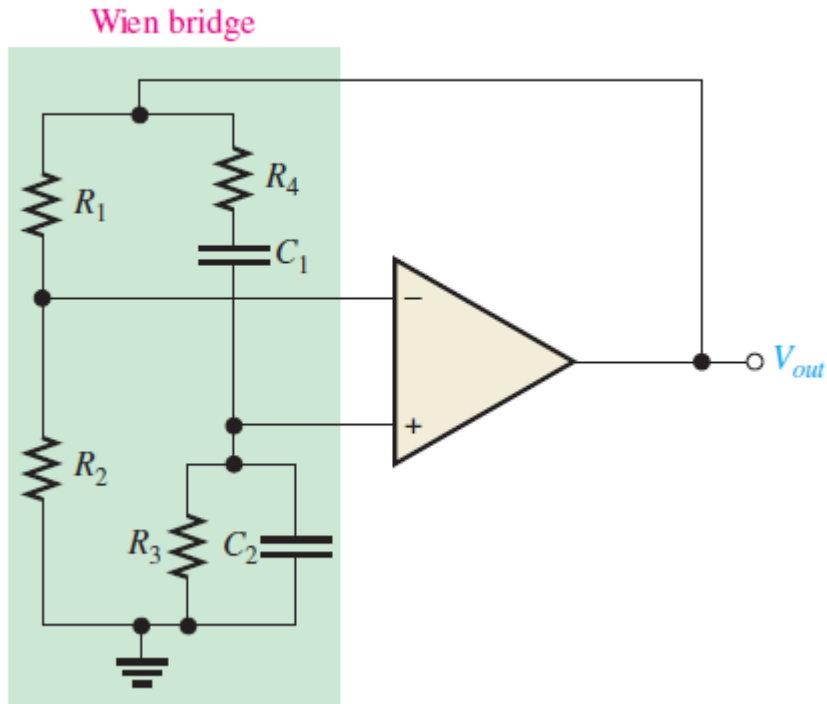
$$\frac{V_{out}}{V_{in}} = \frac{R(-jX)/(R - jX)}{(R - jX) + R(-jX)/(R - jX)} = \frac{R(-jX)}{(R - jX)^2 - jRX}$$

$$f_r = \frac{1}{2\pi RC}$$

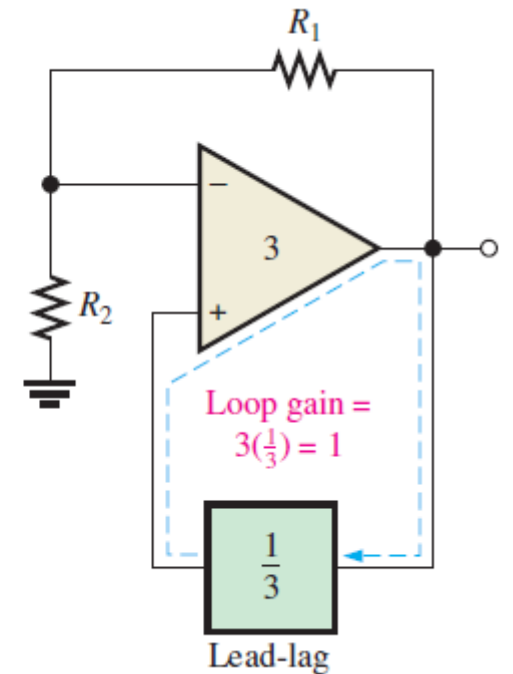
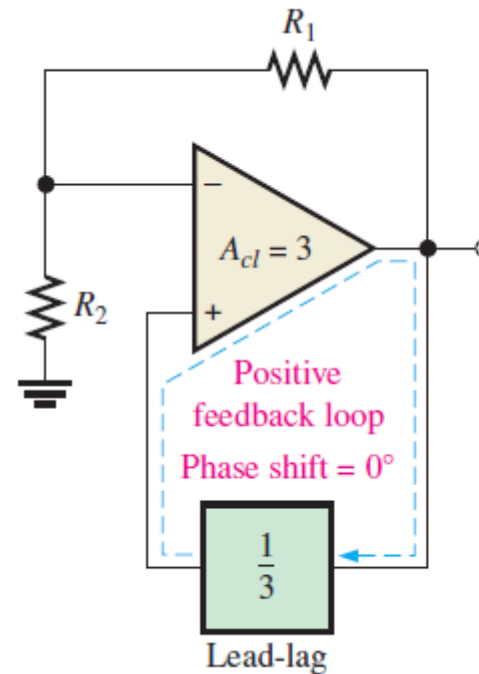
$$\frac{V_{out}}{V_{in}} = \frac{1}{3}$$



Outcome 1: To understand basics of OP AMPs

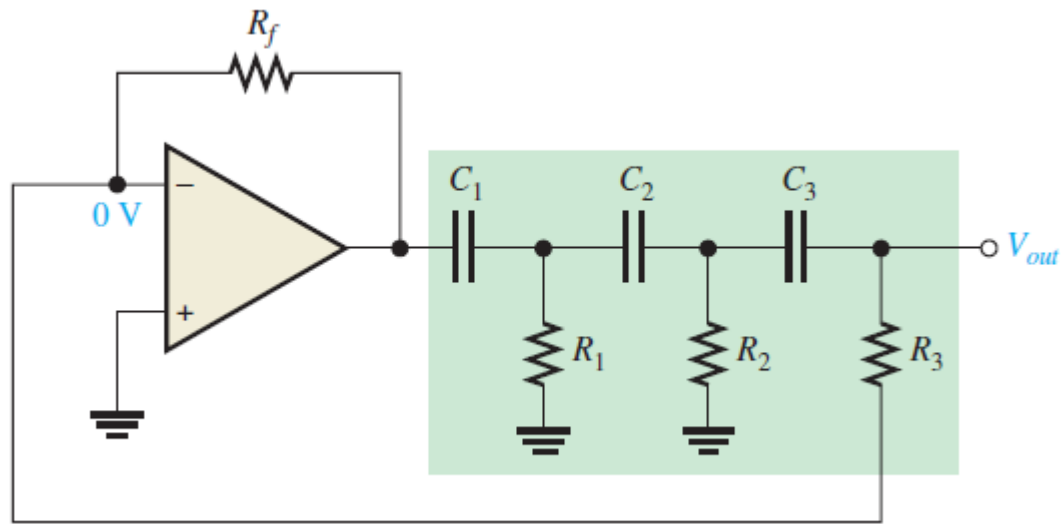


$$A_{cl} = \frac{1}{B} = \frac{1}{R_2/(R_1 + R_2)} = \frac{R_1 + R_2}{R_2}$$



Outcome 1: To understand basics of OP AMPs

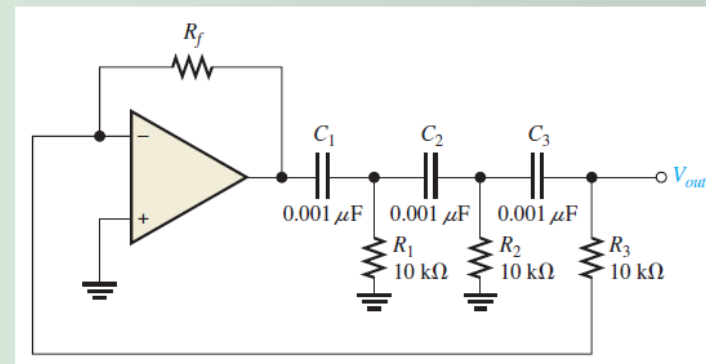
- The Phase-Shift Oscillator



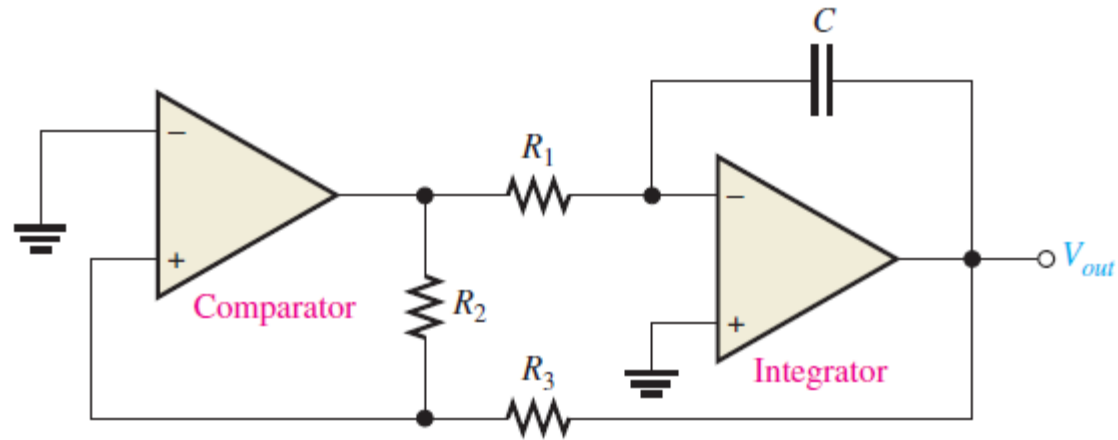
$$B = \frac{1}{29}$$

$$f_r = \frac{1}{2\pi\sqrt{6RC}}$$

- (a) Determine the value of R_f necessary for the circuit in Figure 16-14 to operate as an oscillator.
- (b) Determine the frequency of oscillation.

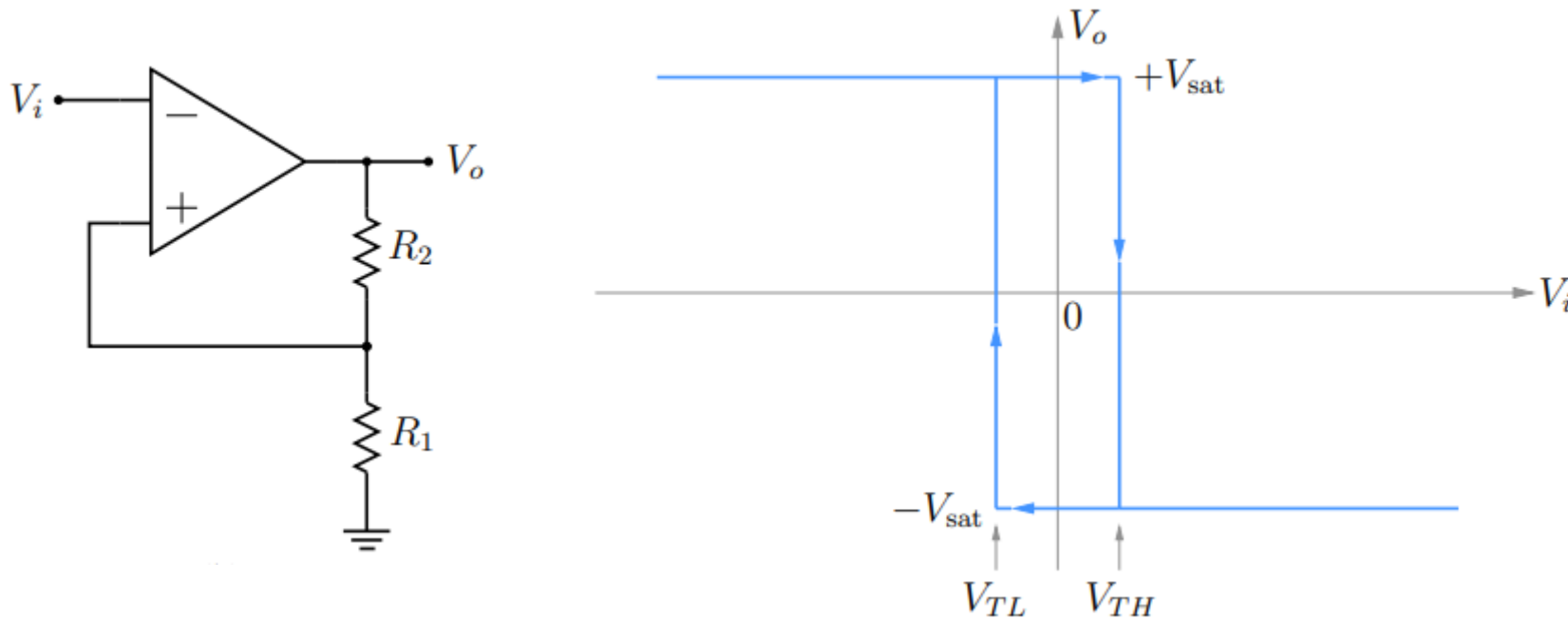


Outcome 1: To understand basics of OP AMPs



Outcome 1: To understand basics of OP AMPs

- Schmitt Trigger

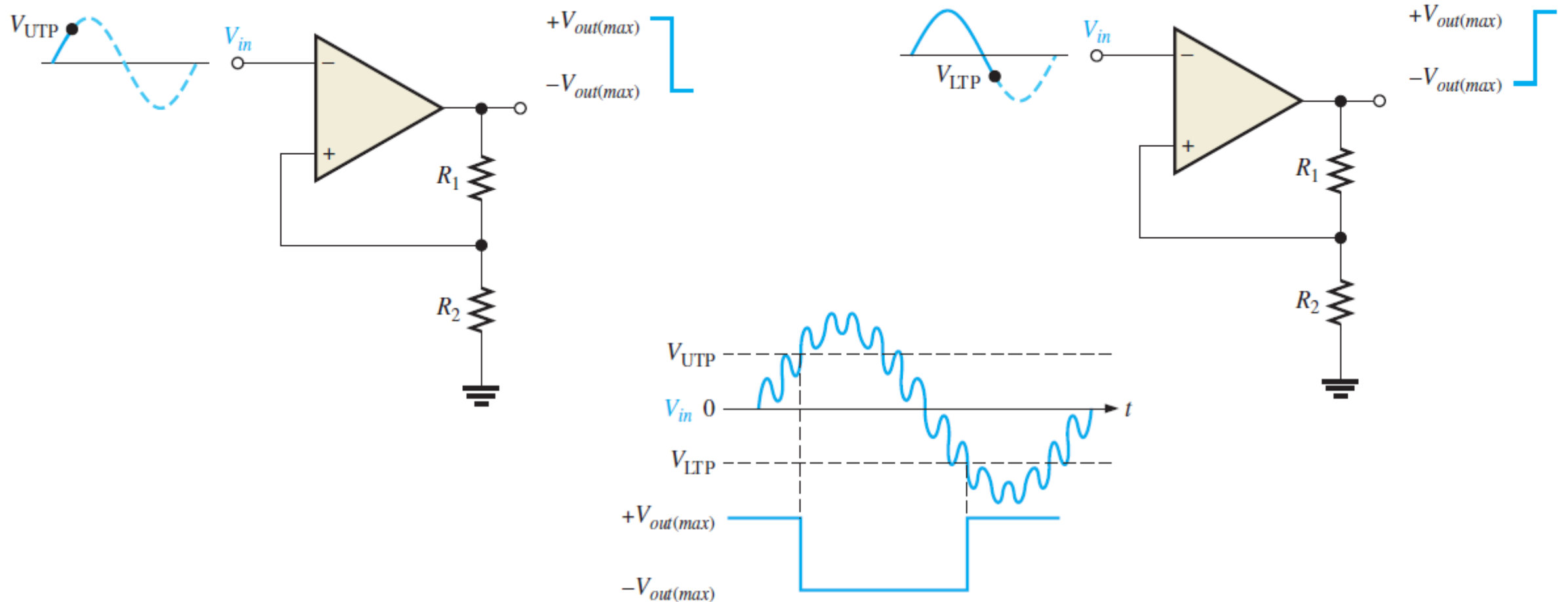


$$V_{UTP} = \frac{R_2}{R_1 + R_2} (+V_{out(max)})$$

$$V_{LTP} = \frac{R_2}{R_1 + R_2} (-V_{out(max)})$$

$$V_o = A_v (V_+ - V_i)$$

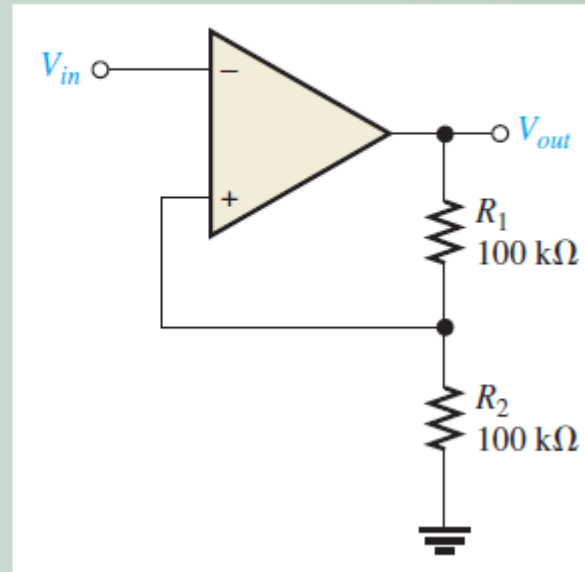
Outcome 1: To understand basics of OP AMPs



Outcome 3

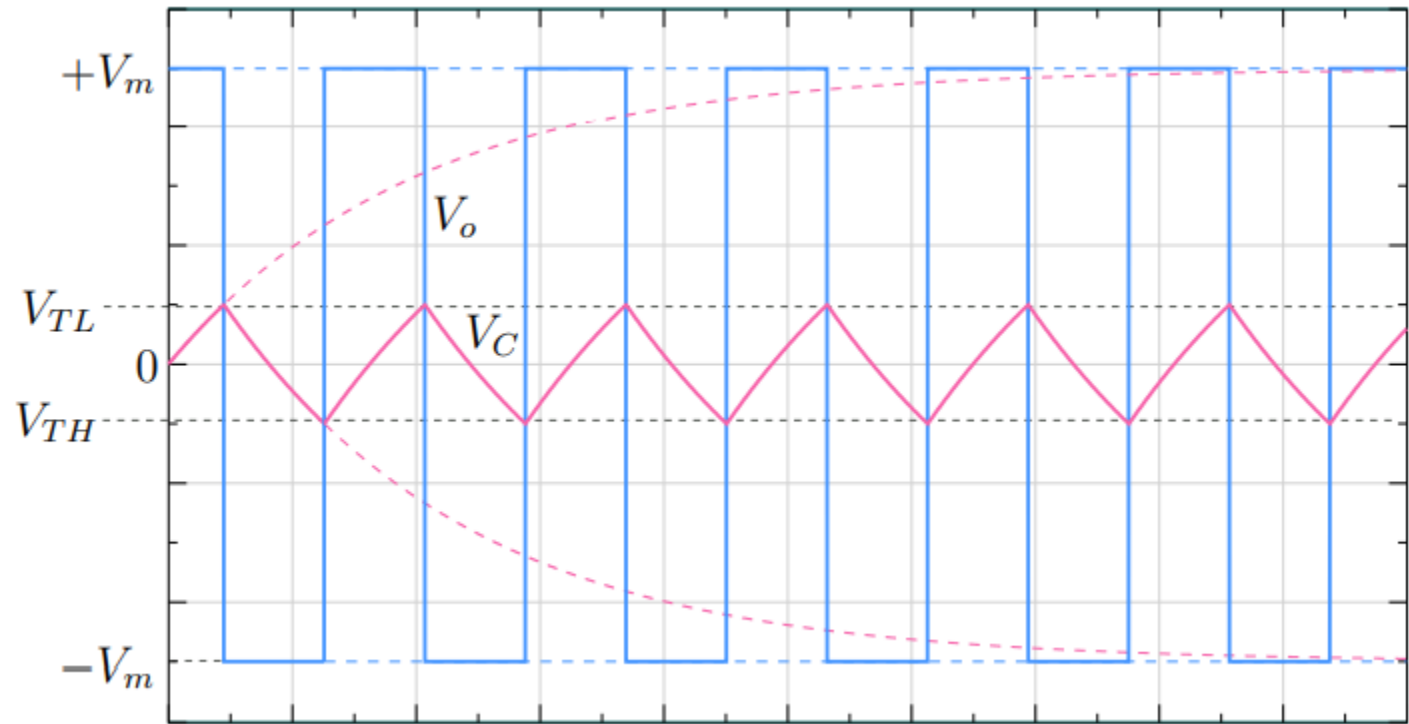
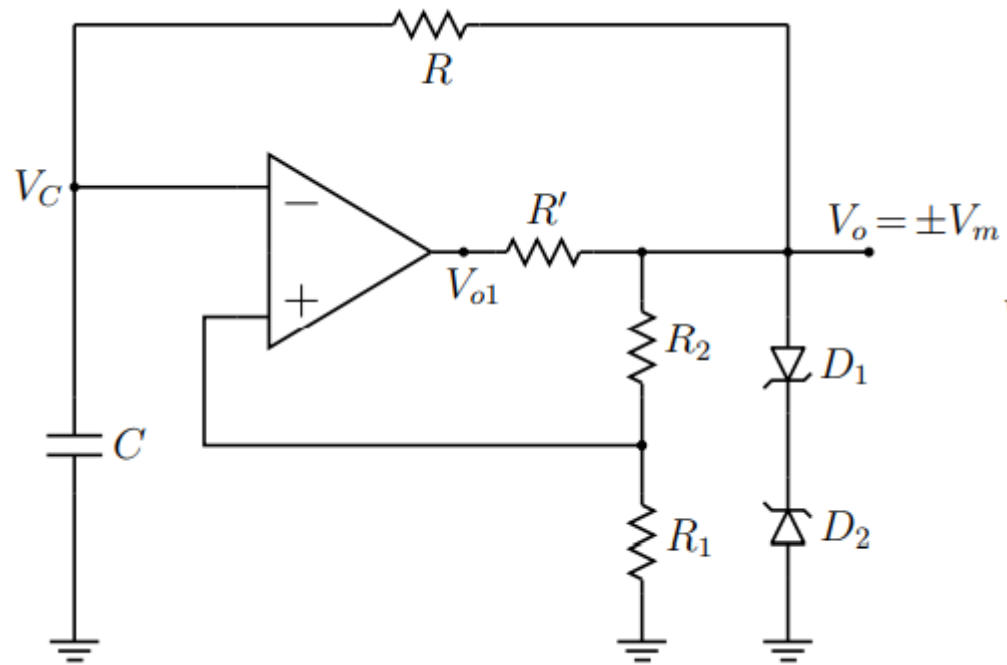
Determine the upper and lower trigger points for the comparator circuit in Figure 13–9. Assume that $+V_{out(max)} = +5\text{ V}$ and $-V_{out(max)} = -5\text{ V}$.

► FIGURE 13–9



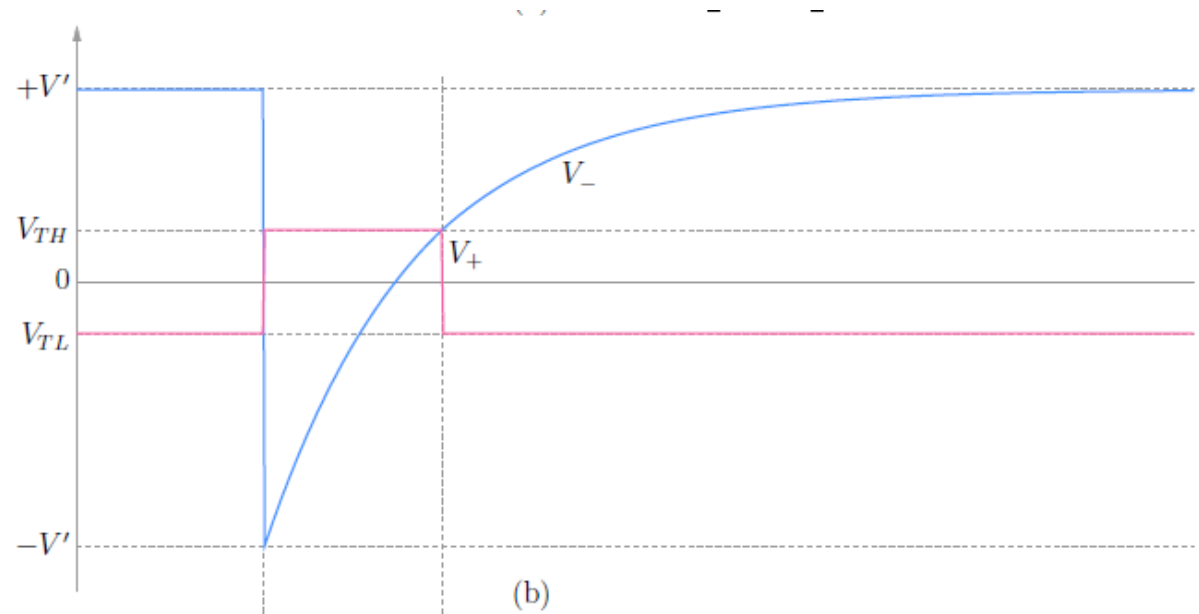
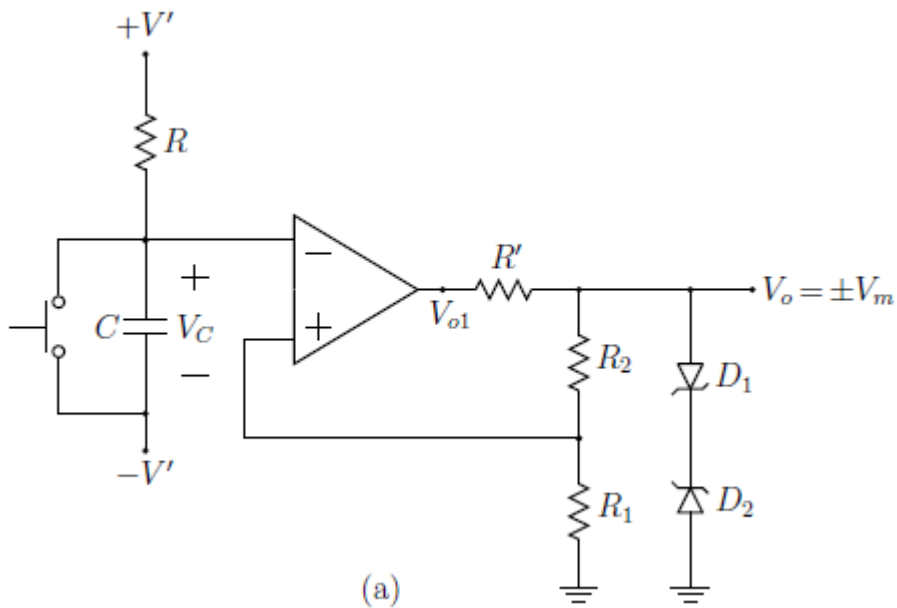
Outcome 1: To understand basics of OP AMPs

Astable multivibrator



Outcome 1: To understand basics of OP AMPs

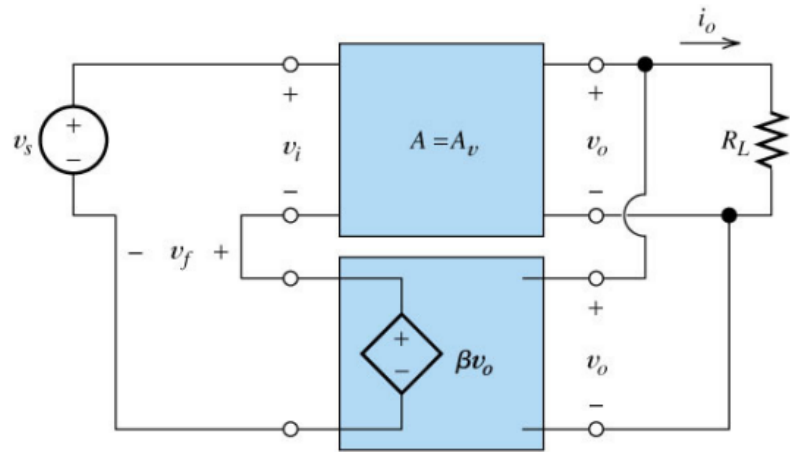
Monostable multivibrator



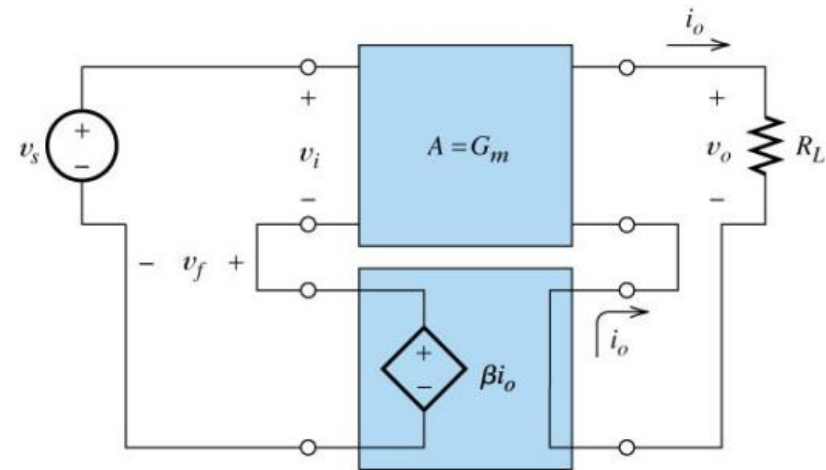
Outcome 1: To understand basics of OP AMPs

- Basic Feedback
- Depending on the input signal (voltage or current) to be amplified and form of the output (voltage or current), amplifiers can be classified into four categories. Depending on the amplifier category, one of four types of feedback structures should be used.
- Voltage series feedback ($A_f = V_o/V_s$) – Voltage amplifier
- Voltage shunt feedback ($A_f = V_o/I_s$) – Trans-resistance amplifier
- Current series feedback ($A_f = I_o/V_s$) - Trans-conductance amplifier
- Current shunt feedback ($A_f = I_o/I_s$) - Current amplifier Topologies

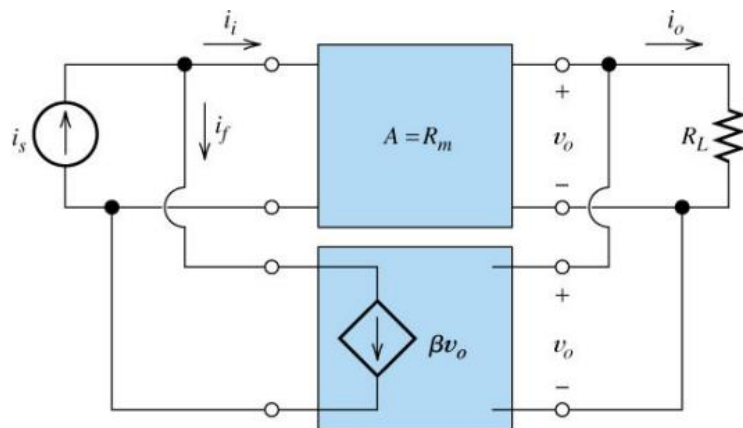
Outcome 1: To understand basics of OP AMPs



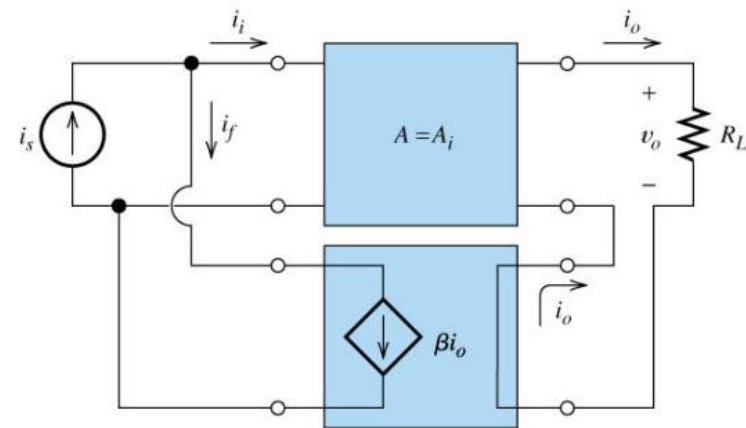
(a) Series voltage feedback



(b) Series current feedback



(c) Parallel voltage feedback



(d) Parallel current feedback